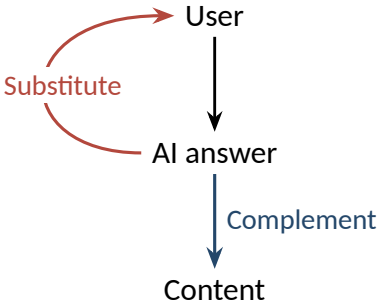
Artificial Intelligence and the Declining Knowledge Commons

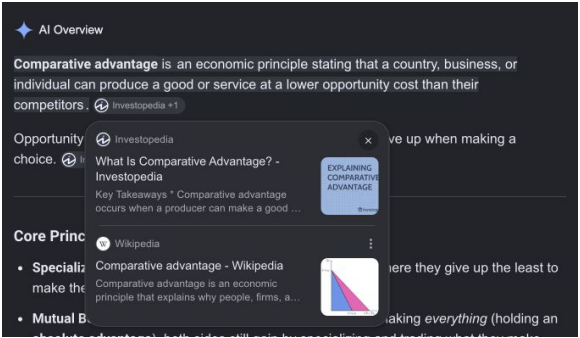
Robert Leung

How will AI disrupt the market for search?

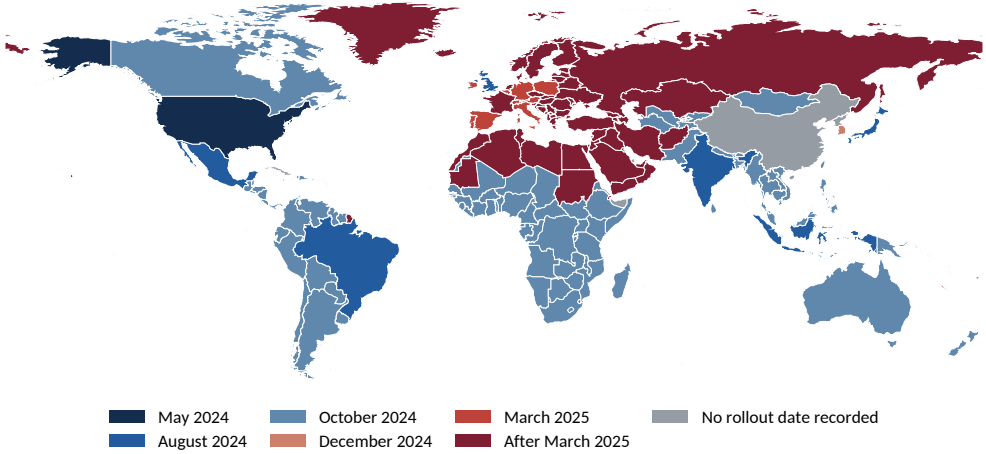
AI as a new good



“What is comparative advantage?”



Setting: AI Overviews and Wikipedia



Research questions

How does AI search change the production and circulation of online knowledge?

1. **Consumption:** Does AI replace visits to original sources?

2. **Composition:** Which kinds of knowledge lose readers?

3. **Production:** Does losing readers reduce contributions and maintenance?

Research questions

How does AI search change the production and circulation of online knowledge?

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   March cohort: ∼3% fewer Google search referrals.
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Research questions

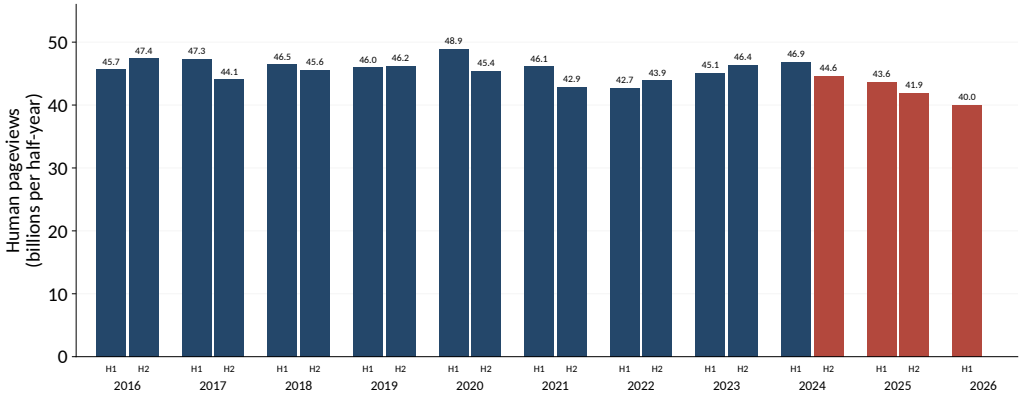
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   Articles read more heavily in March AIO countries see ∼4% fewer contributing editors in April.

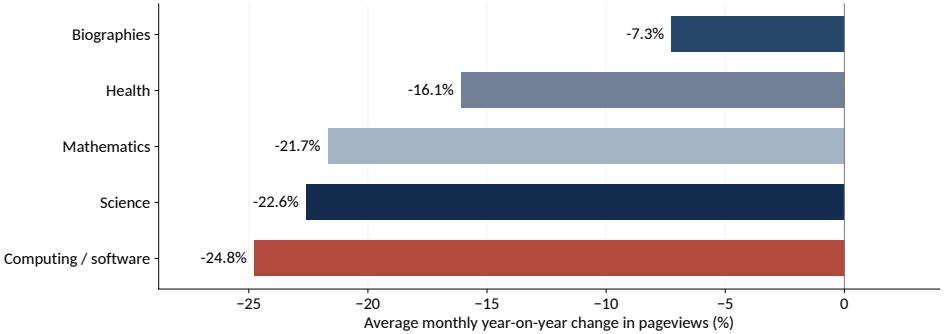
Related literature

- **Khosravi and Yoganarasimhan (2026):** English versus German/French versions of the same articles; rollout DiD. Wikipedia search referrals fall 4.8–5.5%.
- **Agarwal and Sen (2026):** Randomized removal of AIOs using a browser extension. AIOs reduce organic clicks by 39.8% when they appear.
- **Wang et al. (2026):** Randomized Google Search interfaces. Removing AI features raises external click-through by 8.8 percentage points.
- **Lyu et al. (2025):** DiD by article overlap with ChatGPT around its launch. Newly created, popular overlapping articles lose more views and edits.
- **Reeves, Yin, and Simperl (2025):** Pre/post panel of 12 language editions. No overall engagement decline after ChatGPT; slower growth in supported languages.

English Wikipedia readership has declined



Traffic has declined differentially

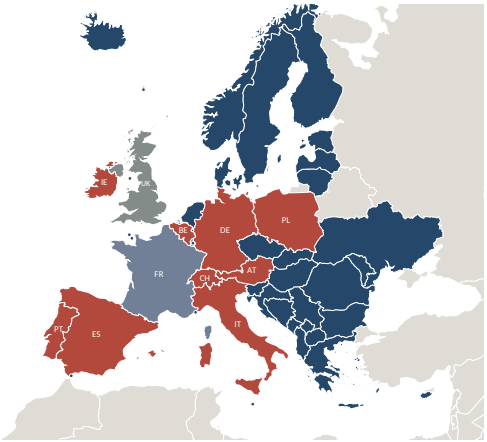


But how much of this is attributable to AI?

Identification strategies

1. March EU rollout
2. Language editions
3. Search Exposure
4. Content restrictions

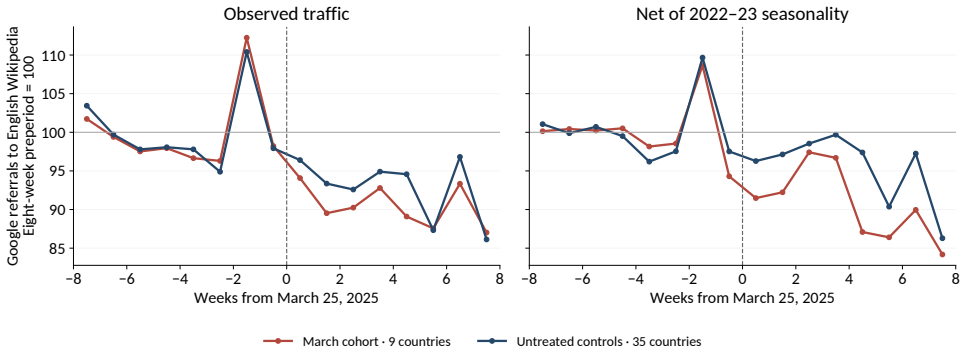
March rollout



9 treated countries
35 comparison countries

- 25 Mar 2025: 9 treated
- 20 May 2025: 34 controls
- 22 Jul 2026: France
- 15 Aug 2024: UK excluded

March cohort: treated and untreated traffic



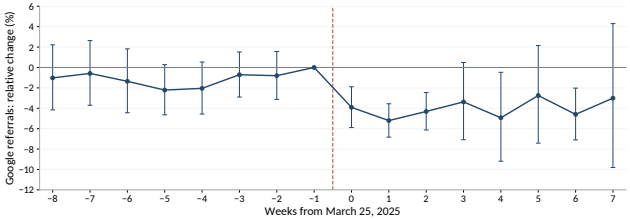
Specifications

PPML event study

$$\mathbb{E}[G_{ct} \mid c, t] = \exp\left(\alpha_c + \delta_t + \sum_{k \neq -1} \beta_k \, \mathrm{March}_c \, 1[w(t) = k]\right)$$

Term	Definition
G_{ct}	Google referrals to English Wikipedia in country c on day t
α_c	Country fixed effects
δ_t	Calendar-date fixed effects
March_c	1 for the nine March cohort countries; 0 for controls
$w(t),\ k$	Event week relative to March 25, 2025; week -1 omitted
β_k	Treated–control difference in week k, relative to week -1

March rollout: PPML DiD



- PPML
  $-2.98\%^{*}$ (SE: 1.29)
- Log DiD
  $-2.17\%^{*}$ (SE: 1.07)
- PPML, seasonal adjustment
  $-3.25\%^{**}$ (SE: 1.29)
- Log DiD, seasonal adjustment
  $-4.54\%^{**}$ (SE: 1.38)

$^{*}p < .10$; $^{**}p < .05$; $^{***}p < .01$

Placebos Seasonality Country weights Staggered DiD

Language editions

Compare English with editions not yet covered by AIO, relative to the same edition pairs in untreated countries.

October: 14 countries Benin, Burkina Faso, Cameroon, Canada, Côte d'Ivoire, Haiti, Kyrgyzstan, Madagascar, Mali, Mauritius, Senegal, Singapore, Suriname, Togo

French, Russian, Chinese or Dutch controls

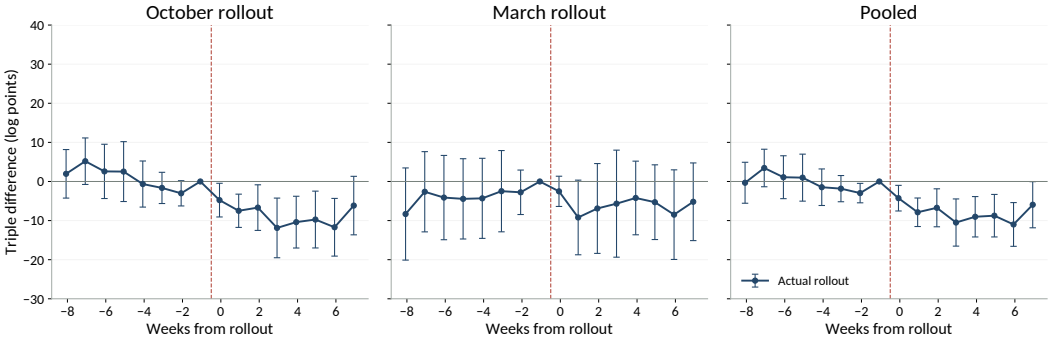
March: 4 countries Austria: Croatian

Belgium: Dutch, French, German

Italy: French

Poland: German

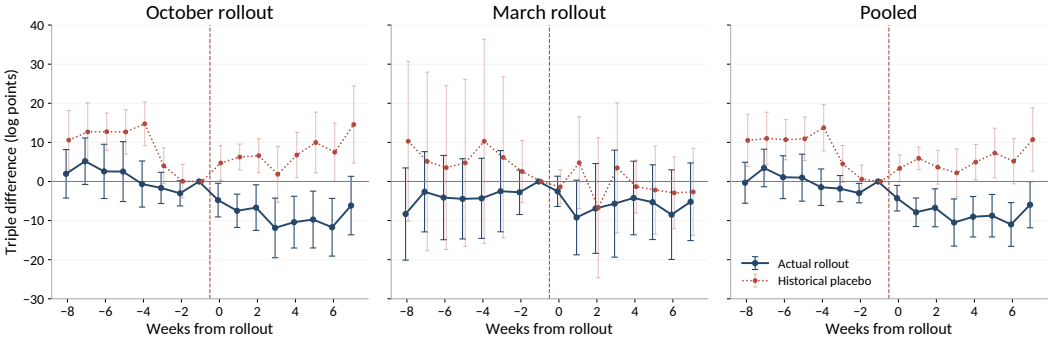
Language editions: triple difference



October: -9.47 (SE: 1.78) March: -2.30 (SE: 2.50) Pooled: -7.87 (SE: 1.44) log points

Within-country DiDs Staggered within-country DiD

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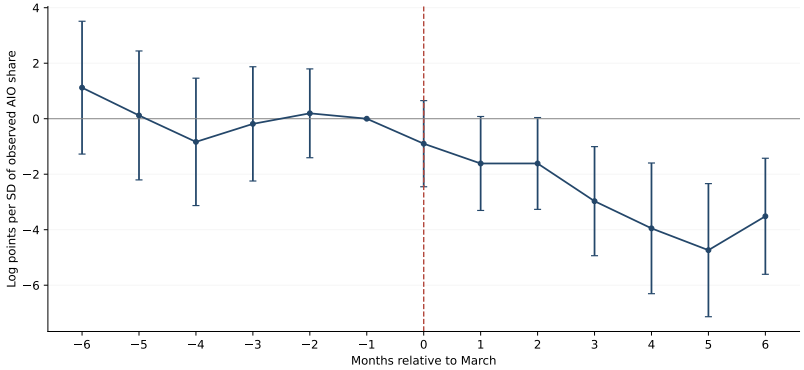
Search Exposure

1. **Historical searches:** ORCAS links Bing queries to the Wikipedia articles users clicked.
2. **Observed AIOs:** Match those queries to a Google benchmark recording AIO appearance in December 2025.
3. **Article exposure:** Share of matched queries that generate an AIO.

1,686 articles; 97% have one matched benchmark query.

Do more-exposed articles lose more search traffic around March?

Search Exposure: March



March–April versus December–February: -1.26 (SE: 0.86) log points per SD.

Placebo years

Search Exposure: an example

Property tax

ORCAS query	AIO in December 2025 benchmark
house taxes property tax	Yes
property taxes	No

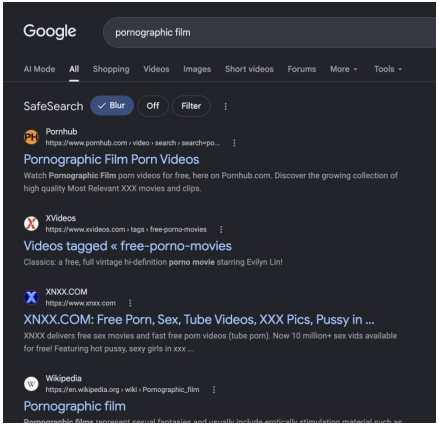
Exposure: 1 of 2 tested queries produces an AIO $= 50\%$.

ORCAS links 41 queries to this article; 39 were not tested.

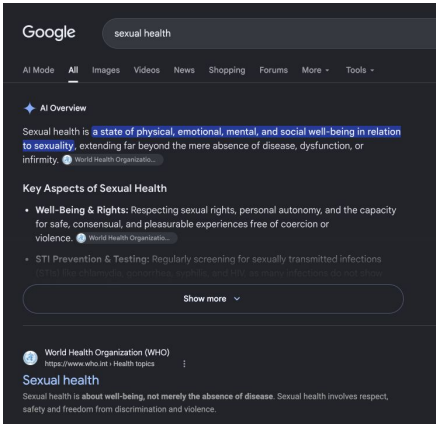
All 41 ORCAS queries

Content restrictions

Explicitly sexual content: no AIO



Sexual health: AIO



Content restrictions: specification

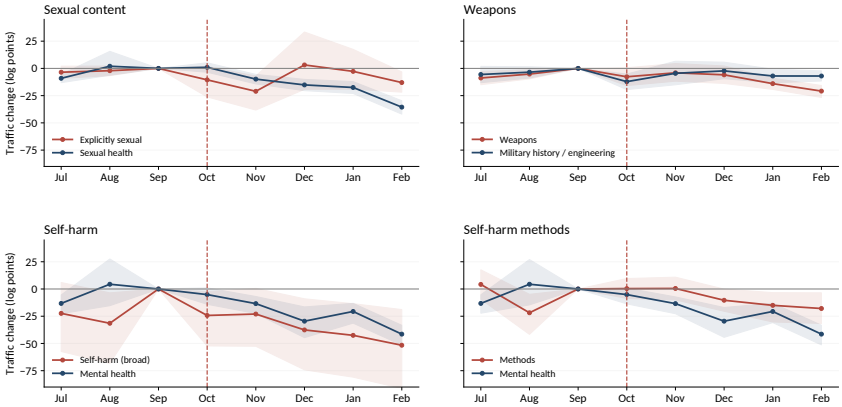
October 2024: compare less-restricted and restricted content

$$y_{gt} = \alpha_g + \delta_t + \beta \, \text{Comparison}_g \times \text{Post}_t + u_{gt}$$

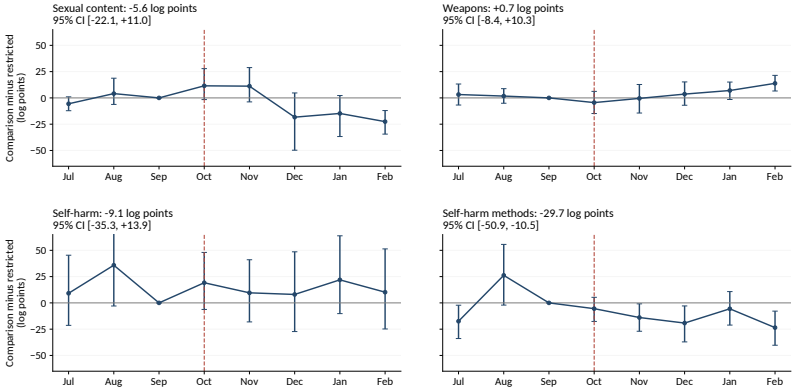
Boundary	Restricted group	Comparison group
Sexual content	Explicitly sexual (592)	Sexual health (1,149)
Weapons	Weapons (4,907)	Military history / engineering (3,656)
Self-harm	Broad self-harm (220)	Mental health (1,152)
Self-harm methods	Methods subset (47)	Mental health (1,152)

y_{gt}: log worldwide search referrals, net of the mean 2022/23 seasonal path.
Post: November–December; pre: August–September.
Estimate separately by boundary. Negative β: comparison falls more.

Content restrictions: traffic paths

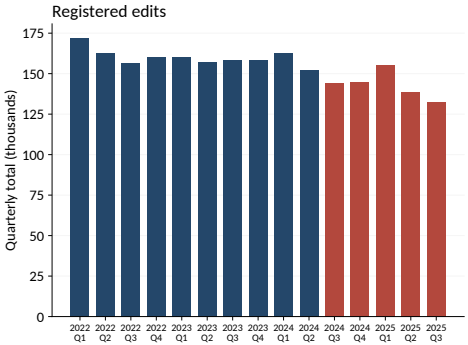


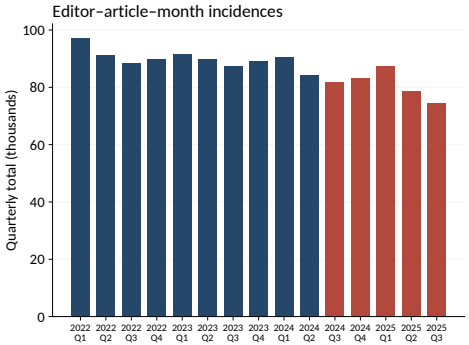
Content restrictions: differences



Placebos May launch Matched sexual-content sample

Supply side experiments





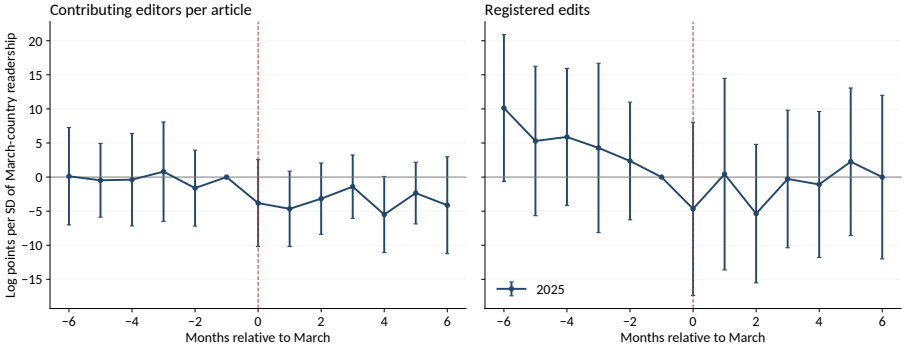
Supply side experiments: specification

PPML on monthly editor and edit counts: 2,872 articles

$$\mathbb{E}[Y_{it}] = d_t \exp\left[\alpha_i + \lambda_{q(i),t} + \sum_{k \neq -1} \left(\beta_k D_i + X_i' \theta_k\right) 1[k(t) = k]\right]$$

Y_{it}	Registered edits or contributing editors on article i
D_i	Pre-AIO share of readers from March countries, per SD (2.37 pp)
$\alpha_i,\ \lambda_{q(i),t}$	Article and topic-by-month fixed effects
X_i	Other cohort reader shares and 2023 popularity, interacted with month
d_t	Days in month
β_k	Differential response per SD of March-country readership

Supply side experiments: PPML

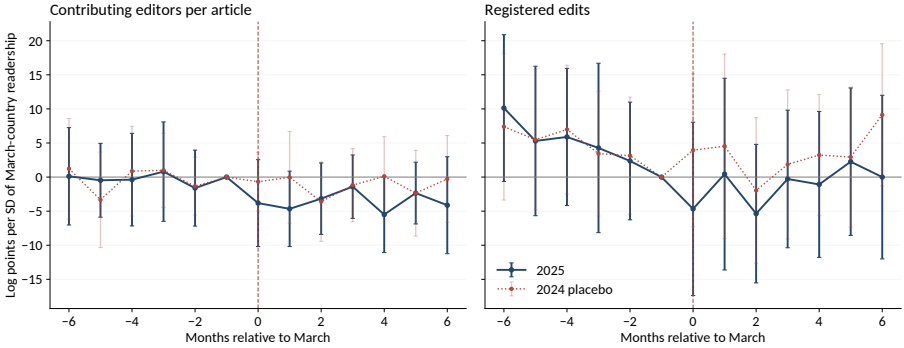


April versus Jan–Feb: editors -3.85 (SE: 2.06); edits -0.75 (SE: 6.49).

Log points per SD of pre-rollout March-country readership.

Seasonal baselines Placebo years Log-count model

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Log points per SD of pre-rollout March-country readership.

Seasonal baselines Placebo years Log-count model

Next steps

1. **Better data:** Ideal to have individual search data (e.g. Sensor Tower, Comscore, etc.)

2. **Beyond Wikipedia:** Paid content (e.g. Substacks, news, etc.)

3. **Monetization:** Impact on ad revenue